

⌈ Preliminary.

⌈  $(\mathbb{Z}_n, +, \cdot)$  is a Ring. If  $n=p$  is prime, then for any  $a \in \mathbb{Z}_p$ , because

$$a^{p-1} \equiv 1 \pmod{p}.$$

That is,  $a^{-1} = a^{p-2} \pmod{p} \in \mathbb{Z}_p$ .

⌈ Let  $F$  be a field. If  $\text{ch}(F)=0$ , then  $F$  must contain  $\mathbb{Q}$ ,

If  $\text{ch}(F) \neq 0$ , then  $F$  must contain  $\mathbb{F}_p$ , where  $p = \text{ch}(F)$

⌈ If  $\deg_{\mathbb{F}_p} \alpha = n$ , then  $F(\alpha)$  has  $p^n$  elements.

Because  $F(\alpha) \cong \mathbb{F}_p^n$

⌈ If  $E$  is finite field, then  $E$  has  $p^n$  elements for some  $n \in \mathbb{N}$ , where  $p = \text{ch}(E)$ .

Because since  $E$  is finite,  $\dim_{\mathbb{F}_p} E$  must be finite.

⌈ Thm. Let  $\mathbb{Z}_p \subseteq E \subseteq \overline{\mathbb{Z}_p}$  has  $p^n$  elements. Then, for  $p(x) = x^{p^n} - x \in \mathbb{Z}[x]$ ,

$$E = \{\alpha \in \overline{\mathbb{Z}_p} \mid p(\alpha) = 0\}$$

Proof. Since  $(E \setminus \{0\}, \cdot)$  is a group of order  $p^n - 1$ , for  $\alpha \in E \setminus \{0\}$ ,  $|\alpha|$  divides  $p^n - 1$ .

So that  $\alpha^{p^n - 1} = 1$ , therefore  $p(\alpha) = 0$ . And, the polynomial  $p(x)$  can have at most  $p^n$  zeros, consequently we obtain the result.

Thm. Let  $F$  be a finite field of characteristic  $p$ .

Then, a map

$$\varphi_p: F \rightarrow F: x \mapsto x^p$$

is Automorphism. Moreover,  $F_{\{\varphi_p\}} = \mathbb{Z}_p$ .

Proof. Well-defined is clear.

Homomorphism: for any  $a, b \in F$ ,

$$\varphi_p(a+b) = (a+b)^p = \underbrace{\sum_{k=0}^p \binom{p}{k} a^{p-k} b^k}_{\text{binomial}} = a^p + b^p = \underbrace{\varphi_p(a)}_{\text{ch}(F)=p} + \varphi_p(b)$$

And clearly preserves multiply.

Injectivity:

$$\varphi_p(a) = a^p = 0 \Leftrightarrow a = 0 \Rightarrow \ker \varphi_p = \{0\}.$$

Surjectivity: Finite injective endomorphism, thus surjective.

Meanwhile, by the Fermat's Theorem, for any  $a \in \mathbb{Z}_p$ , (isomorphically the prime field  $\mathbb{Z}_p$ )

$$\varphi_p(a) = a^p = a$$

Thus  $\mathbb{Z}_p \subseteq F$ . And, observe that

$$\varphi_p(x) = x \Leftrightarrow x^p = x \Leftrightarrow x^p - x = 0.$$

This polynomial has at most  $p$  zeros in  $F$ , thus  $\mathbb{Z}_p = F_{\{\varphi_p\}}$ .

Corollary. If  $\text{ch}(F) = p$ , then  $(\alpha + \beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$  for any  $\alpha, \beta \in F$ ,  $n \geq 1$ .

The existence of  $GF(p^n)$ .

Thm. If  $F$  is a field with  $\text{ch}(F)=p$ , then  $x^{p^n}-x \in F[x]$  has  $p^n$  distinct zeros in  $\overline{F}$ .

Proof. observe that  $x^{p^n}-x = x \cdot [x^{p^n-1}-1]$ . Let  $f(x) = x^{p^n-1}-1$ .

Suppose that  $\alpha \in \overline{F}$  is a root of  $f(x)$ . Then

$$f(x) = (x-\alpha) \cdot \underbrace{(x^{p^n-2} + \alpha x^{p^n-3} + \dots + \alpha^{p^n-3}x + \alpha^{p^n-2})}_{p^n-1 \text{ terms}}$$

Let  $g(x) = f(x)/(x-\alpha)$ . Now,

$$g(\alpha) = [(p^n-1) \cdot 1] \cdot \alpha^{-1} = -\alpha^{-1} \quad (\text{since } (p \cdot 1)x = 0, \text{ ch}(F)=p)$$

But  $\alpha \neq 0$ , thus  $-\alpha^{-1} \neq 0$ . This means  $\alpha$  is a zero of  $f$  of multiplicity 1.

A finite field of order of  $p^n$  exists.

Proof. Let  $K = \{x \in \overline{\mathbb{Z}_p} \mid x^{p^n}-x=0\} \subseteq \overline{\mathbb{Z}_p}$